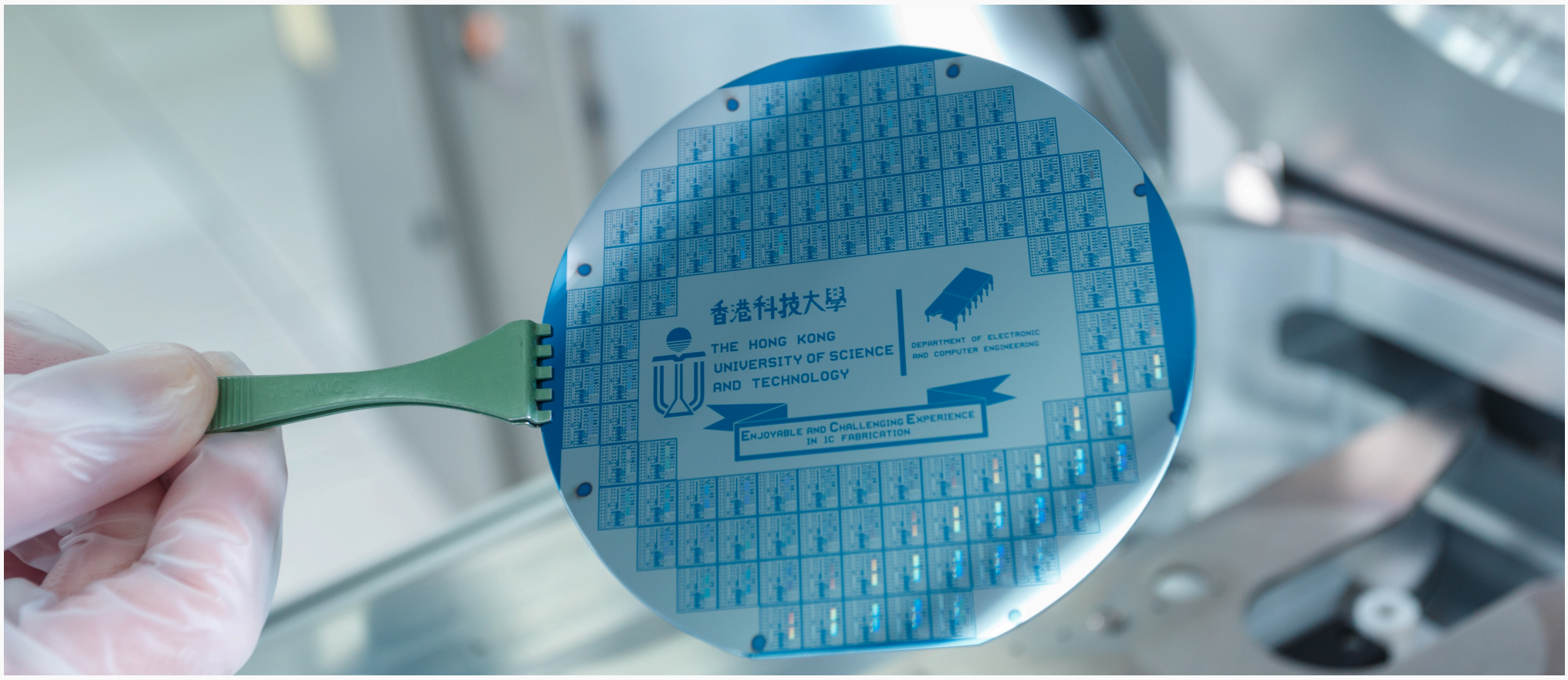


BENG IN MICROELECTRONICS AND INTEGRATED CIRCUITS

Degree Awarded: BEng in Microelectronics and Integrated Circuits

JUPAS Code: JS5250 (through admission to the Department of Electronic and Computer Engineering)

Abbreviated Title: BEng (MEIC)

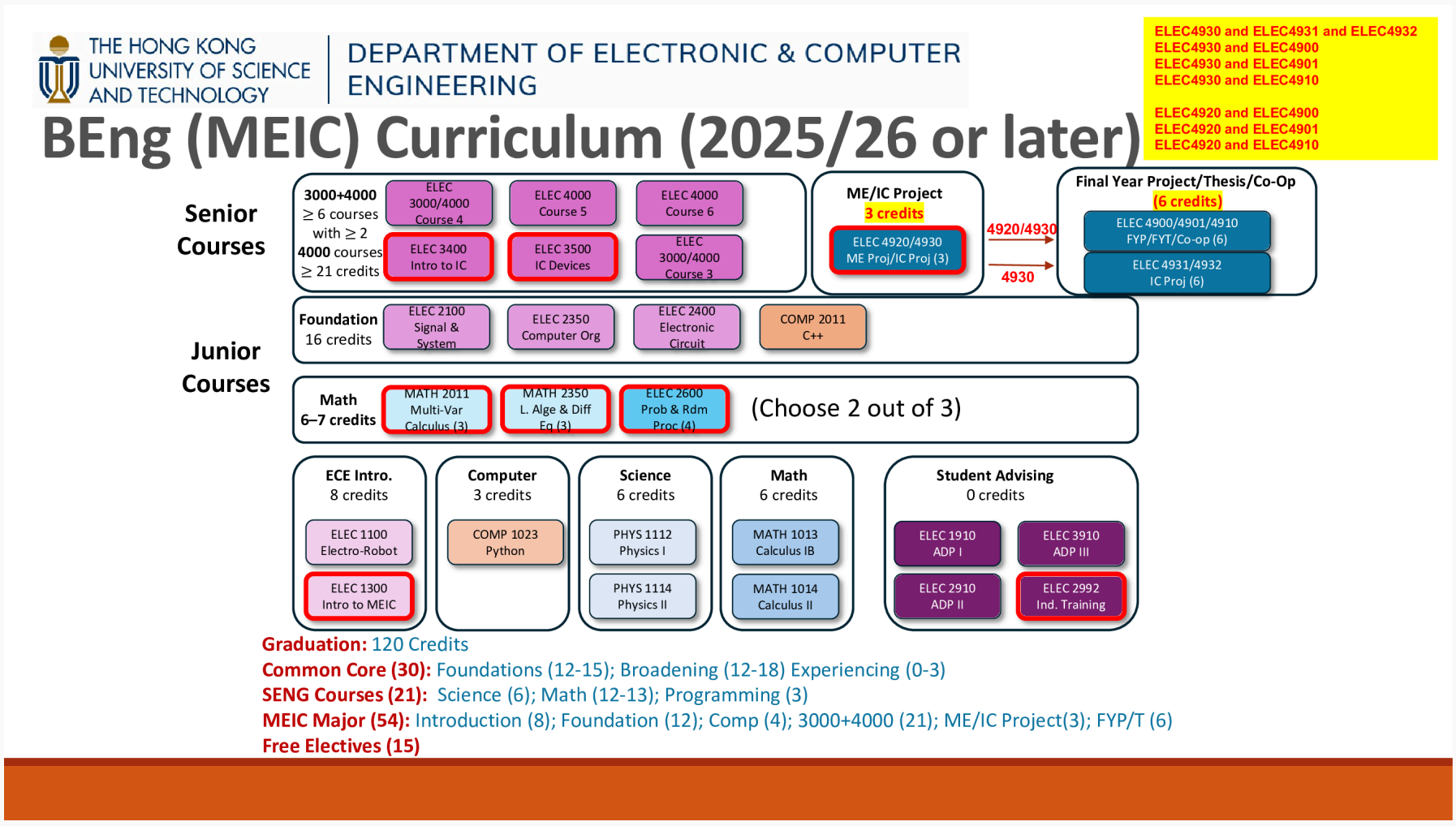
Program Duration: Normally 4 years

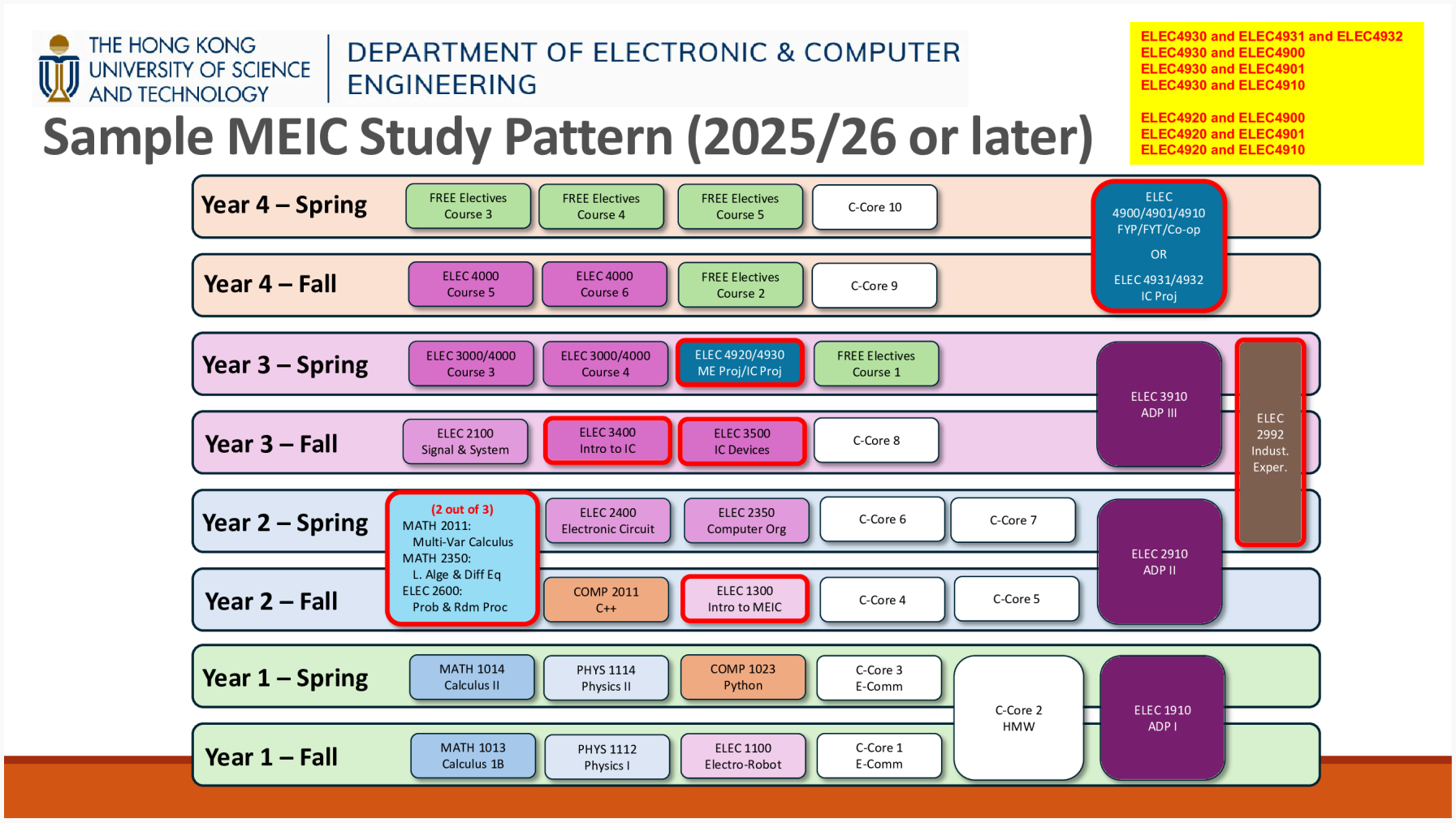
Programme Outcomes(PO)

PO1	An ability to apply knowledge of mathematics, science, microelectronics, and integrated circuits.
PO2	An ability to design and conduct experiments, as well as to analyze and interpret data.
PO3	An ability to design efficient and economical microelectronics systems and integrated circuits, components, or processes subject to practical constraints.
PO4	An ability to function in a multi-disciplinary environment through teamwork.
PO5	An ability to identify, formulate and solve microelectronics and integrated circuits problems.
PO6	An ability to understand professional practices and ethical responsibilities.

PO7	An ability to communicate effectively.
PO8	An ability to understand contemporary global, regional, economic, environmental, and social issues, and the corresponding role and the impact of microelectronics and integrated circuits design engineers.
PO9	An ability to recognize the need for, and to engage in life-long learning.
PO10	An ability to use current techniques, skills, and engineering tools necessary for solving microelectronics and integrated circuits problems.
PO11	An ability to use the computer/IT tools relevant to microelectronics systems and integrated circuits design along with an understanding of their processes and limitations.

The Curriculum





Research Option

Students may opt to graduate with or without an option. Students who take the Research Option must complete all requirements specified in addition to the major requirements.

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Prerequisite

Students may refer to the latest course catalog to review the prerequisite requirements of the senior courses of interest and plan their course selection accordingly.

Critical Milestones

Students should follow critical milestones to ensure that they could graduate on schedule.

2018/19 or later

University Core and Required Courses

Introductory courses show students the big picture of ECE through two lab-based courses that all students must take early in their studies. Foundation courses provide students the core knowledge of ECE through four courses that all students must take as prerequisites for higher-level courses. Science / Math courses provide essential background knowledge to prepare students for higher-level courses. All students are required to take two Science courses, two Calculus courses, and three out of four other Math courses.

University Core (2022/23 or after)

University Core (2021/22 or before)

Introductory / Foundation Courses

Science / Math / Comp Courses

Senior Courses

Senior courses broaden students' horizon and equip them with the necessary tools in different ECE areas, and allow students to deepen their knowledge within a selected area of interest. Students are required to take at least 21 credits of senior courses including at least two ELEC 4000-level courses.

Artificial Intelligence and Data Science

Bioelectroincs and Bioinformation Technology

Control and Robotic Systems

Photonics and Microwave Technology

Embedded Systems

Integrated Circuits and Systems

Semiconductor Devices and Technology

Signal Processing and Communications

Recommended Study Patterns (To be provided)

The recommended study patterns help students plan their studies in selected areas.

Microelectronics

Integrated Circuits

Curriculum of Other Years of Entry

Students must adhere to the curriculum of the year of entry throughout their program of study. Details of the degree and major requirements can be found in the program catalog of the year of intake.

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Program Requirements

Please refer to the program catalog for the latest information

Final Year Projects/Theses

All ECE students are required to complete a Final Year Project/Thesis (FYP/T). It is an excellent opportunity for them to synthesize the theory and experimental work they learned during the Program in order to complete the project.

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Industrial Experience

All ECE students are required to undertake practical training as a compulsory graduation requirement. Students could gain valuable hands-on experience through training and enhance their understanding and appreciation of the knowledge acquired in classrooms and laboratories. To satisfy the practical training requirement, students may either complete an internship or the required Industrial Training modules.

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Minor Program

Undergraduate students may enroll in a minor program in addition to their disciplinary major if they meet the enrollment conditions specified. On successful completion of both major and minor programs, the student will receive a degree in their major subject with a minor in their selected program.

Minor Program in Robotics

Minor Program in Sustainable Energy Engineering

IT Minor (with Computer Science & Engineering Program)

Students may also enroll in minor programs offered by other Schools in the University.

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Double Major

With the approval of the major department, undergraduate students can graduate with dual program designation (e.g. BEng in Electronic Engineering AND Computer Engineering), if they can complete the requirements of the two undergraduate programs within the normal period of study.

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Exchange Programs

The University has established exchange programs with leading institutions, such as UC Berkeley, University of Michigan, UCLA, University of Pennsylvania, Tsinghua University, University of Science and Technology in China, and is in active negotiation with a number of institutions. These exchange programs intend to provide opportunities for students to study abroad and enhance their international exposure and future career development.

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Academic Advising

The ECE Department has set up several channels to help students in academic affairs: student advising courses and FYP supervisors. All students have a mentor for each year of their major study, and they should make use of these chances to ask questions and discuss academic issues (comment on courses, instructors, career planning, further study, etc.).

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Professional Recognition

Our programs are recognized by the Hong Kong Institution of Engineers (HKIE). Graduates from our programs with honors will be deemed to have satisfied the educational requirement for Corporate Membership of HKIE.

Career Prospects

A wide variety of career paths are available to MEIC graduates. Upon graduation, we expect our graduates to take positions in the industry as electronic engineers, communication engineers, system analysts and designers, information technologists, computer project managers and many other technical and managerial positions. A significant portion chooses to continue their study at the postgraduate level. Even graduates who choose not to pursue technical careers find that the engineering training they obtained at HKUST has given them a competitive advantage over their co-workers.

The list of employers includes many world-class local and overseas companies, such as ASM, ASTRI, HKT, Huawei, Lenovo, Microsoft, National Semiconductor, Philips, Smartone, Solomon Systech, Varitronix, etc.

Enquiries

Email: eequestions@ust.hk

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